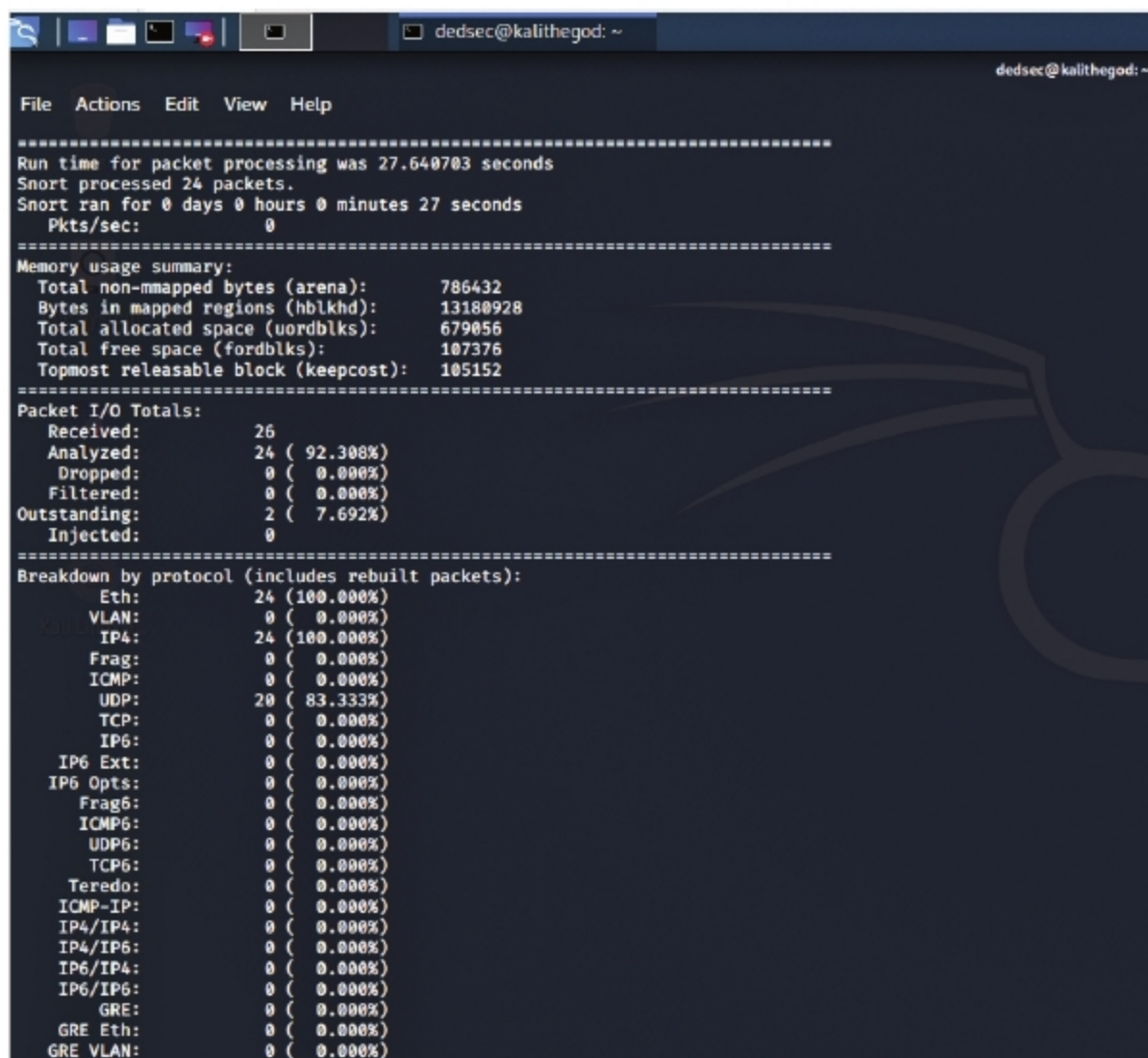


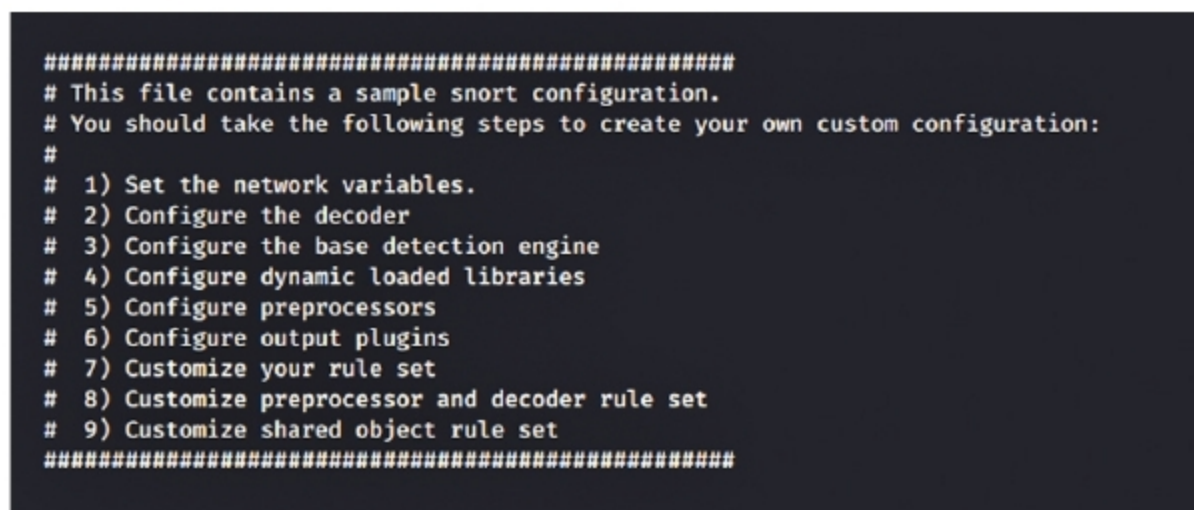


```

File  Actions  Edit  View  Help
=====
Run time for packet processing was 27.640703 seconds
Snort processed 24 packets.
Snort ran for 0 days 0 hours 0 minutes 27 seconds
Pkts/sec: 0
=====
Memory usage summary:
Total non-mapped bytes (arena): 786432
Bytes in mapped regions (hblkhd): 13180928
Total allocated space (uordblks): 679056
Total free space (fordblks): 107376
Topmost releasable block (keepcost): 105152
=====
Packet I/O Totals:
Received: 26
Analyzed: 24 ( 92.308%)
Dropped: 0 ( 0.000%)
Filtered: 0 ( 0.000%)
Outstanding: 2 ( 7.692%)
Injected: 0
=====
Breakdown by protocol (includes rebuilt packets):
Eth: 24 (100.000%)
VLAN: 0 ( 0.000%)
IP4: 24 (100.000%)
Frag: 0 ( 0.000%)
ICMP: 0 ( 0.000%)
UDP: 20 ( 83.333%)
TCP: 0 ( 0.000%)
IP6: 0 ( 0.000%)
IP6 Ext: 0 ( 0.000%)
IP6 Opts: 0 ( 0.000%)
Frag6: 0 ( 0.000%)
ICMP6: 0 ( 0.000%)
UDP6: 0 ( 0.000%)
TCP6: 0 ( 0.000%)
Teredo: 0 ( 0.000%)
ICMP-IP: 0 ( 0.000%)
IP4/IP4: 0 ( 0.000%)
IP4/IP6: 0 ( 0.000%)
IP6/IP4: 0 ( 0.000%)
IP6/IP6: 0 ( 0.000%)
GRE: 0 ( 0.000%)
GRE Eth: 0 ( 0.000%)
GRE VLAN: 0 ( 0.000%)

```

Figure 3: Sniffer output2



```

#####
# This file contains a sample snort configuration.
# You should take the following steps to create your own custom configuration:
#
# 1) Set the network variables.
# 2) Configure the decoder
# 3) Configure the base detection engine
# 4) Configure dynamic loaded libraries
# 5) Configure preprocessors
# 6) Configure output plugins
# 7) Customize your rule set
# 8) Customize preprocessor and decoder rule set
# 9) Customize shared object rule set
#####

```

Figure 4: Conf file sections

Now let us suppose that your machine does not have the required repositories listed, so we may edit them by using the following command:

```
kali > sudo mousepad /etc/apt/sources.list
```

Once done, install Snort. You may verify the installation by running the following command:

```
kali > sudo snort -v
```

As we can see in Figure 1, Snort responds with its currently installed version number.

Basic configuration of Snort

Like all Linux commands and applications, Snort also has the help of the command line, which can be invoked by using the following code:

```
kali > sudo snort --help
```

I have highlighted a few switches from the *Help* section of Snort:

- `-c` gives us the location of the Snort rules and tells it to use its rules. They are the signature against which the new packets are verified.
- `-d` tells Snort to show the application layer of data.
- `-e` displays the Data Link layer of information, which contains the MAC address of the system.
- `-i` allows the user to designate the interface we want to use. By default, Snort uses `eth0`.
- `-k` allows the users to define how they want to store the details of the data capture performed by Snort.
- `-v` is like in most of the programs — verbose, providing all the information.

Starting Snort

Let's get to know some of the basic switches of Snort, by first running `ij`. What makes Snort fabulous is that it can be run as a sniffer, packet logger, or even as a NIDS. In this article, we will look at Snort as a packet sniffer and NIDS.

To run Snort in packet dump mode, use the following command:

```
kali > sudo snort -vde
```

The output we get is pretty self-explanatory (Figures 2). For using Snort as a NIDS, we need to instruct Snort to include the configuration file and rules. Generally, we can find the `conf` file at `/etc/snort/snort.conf` and that file will point to Snort rules. We need to give the `-c` switch and then the location.

```
kali > sudo snort -vde -c /etc/snort/snort.conf
```

We can also customise the rules to suit our needs.

Snort — rules and configuration

Like all general Linux applications, Snort is configured via a `conf` file, which can be opened as a simple text file. Edit this text